tr joelPCM Academic Council

UGANDA ADVANCED CERTIFICATE OF EDUCATION (UACE)

END OF FIRST TERM EXAMINATIONS 2026

S.6 P425/2 PURE MATHEMATICS (Paper 2)

Time Allowed: **2 HOURS 15 MINUTES**

STUDENT NAME:

.....

PERSONAL NUMBER: ..... **SIGNATURE:** .....

DO NOT OPEN THIS BOOKLET UNTIL YOU ARE TOLD TO DO SO BY THE INVIGILATOR

INSTRUCTIONS TO CANDIDATES:

1. Do not write anything in this paper; any rough work should be written in the response booklet(s) provided and crossed out.
2. The time allowed for this paper is strictly **2 Hours 15 Minutes** and no additional time shall be granted.
3. This paper consists of **two** sections; **A** and **B** with a total of **4 items**.
4. Section A has **2** items. Attempt **ALL** items in Section A.
5. Section B has **2** items. Attempt **ONE** item of your choice.
6. **THREE ITEMS** should be attempted in total. Any additional item(s) attempted will not be marked.
7. Begin each item on a fresh page and clearly indicate them in the response booklet provided.
8. Tidy handwriting, meaningful mathematical judgement and conclusion may increase your chances of excelling.
9. Silent non-programmable calculators and logarithmic tables may be used. Take g = 9.8 m/s^2 where applicable.

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SECTION	**A**		**B**		**TOTAL**
ITEM	1	2	3	4	
SCORE					
INITIAL					

SECTION A

Attempt ALL items in this Section

ITEM 1

Mrs. Tendo is a data analyst at the Uganda Bureau of Statistics (UBOS). She is investigating the relationship between monthly rainfall and crop yield across 8 districts of Eastern Uganda for the 2025 agricultural season. The data collected is shown in the table below:

Rainfall, x (mm)	45	60	75	80	95	110	120	130
Crop yield, y (t/ha)	1.2	1.8	2.4	2.6	3.1	3.5	3.8	4.0

In a separate study, UBOS collected data on monthly household incomes of 60 families in Kampala. The data is summarised in the frequency distribution table below:

Monthly income (UGX '000)	200−400	400−600	600−800	800−1000	1000−1200
Frequency	8	15	22	12	3

Mrs. Tendo's major objectives are to assess the strength of the relationship between rainfall and crop yield, predict crop yield from rainfall data, and summarise the household income distribution.

Task:

Help Mrs. Tendo to:

a) (i) Draw a scatter diagram for the rainfall and crop yield data.
   (ii) Calculate the Pearson product-moment correlation coefficient r and comment on the relationship between rainfall and crop yield.

b) (i) Determine the equation of the regression line of y on x in the form $y = a + bx$.
   (ii) Estimate the expected crop yield when the monthly rainfall is 100 mm.

c) Calculate the mean and standard deviation of the monthly household income data.

ITEM 2

Mr. Opio is a quality control manager at a textile factory in Jinja. He uses probability models to monitor production quality and predict outcomes in sampling exercises.

From production records, it is known that 15% of garments produced have minor defects and 5% have major defects. The probability of a garment having both minor and major defects is 0.03. A garment is selected at random from the production line.

In a separate batch quality test, the number of major defects X found per inspection batch follows a Poisson distribution with mean $\lambda = 2$.

The factory also draws a random sample of 10 garments from a large consignment in which 30% are known to be substandard. The random variable Y represents the number of substandard garments found in the sample.

Mr. Opio's major objectives are to determine the probability that a garment is defect-free, predict batch defect rates using the Poisson model, and compute probabilities for the sampling exercise.

Task:

Help Mr. Opio to:

a) Find the probability that a randomly selected garment:
   (i) has a minor or major defect or both

(ii) has only a minor defect but not a major defect

(iii) is completely defect-free

b) Given $X \sim Po(2)$, find:

(i) $P(X = 3)$

(ii) $P(X \leq 2)$

(iii) $P(X \geq 1)$

c) Given $Y \sim B(10, 0.3)$:

(i) Find the expected value $E(Y)$ and the variance $Var(Y)$

(ii) Find $P(Y = 2)$

(iii) Find $P(Y > 8)$

SECTION B

Attempt any ONE item from this Section

ITEM 3

Mr. Kiggundu is a mechanical engineer at a construction company in Kampala. He is analysing the dynamics of two separate systems: a loaded crate on an inclined plane and a collision between two vehicles during a safety test.

A crate of mass 50 kg is pulled from rest up a rough inclined plane that makes an angle of 30° with the horizontal. The rope pulling the crate is parallel to the plane and exerts a tension of 400 N. The coefficient of friction between the crate and the surface of the plane is 0.2.

In a vehicle crash test, a truck of mass 2,000 kg moving at 15 m/s on a straight horizontal road collides directly with a stationary car of mass 800 kg. Immediately after the collision, the truck moves at 8 m/s in its original direction of motion.

Mr. Kiggundu's major objectives are to determine the motion of the crate on the incline and to analyse the collision between the truck and the car.

Task:

Help Mr. Kiggundu to:

a) Find the acceleration of the crate as it moves up the inclined plane.

b) Find the velocity of the crate after it has travelled a distance of 5 m up the plane.

c) (i) Use the principle of conservation of momentum to find the velocity of the car immediately after the collision.

(ii) Calculate the coefficient of restitution and determine whether the collision is elastic or inelastic.

ITEM 4

Ms. Birungi is a sports science analyst studying the dynamics of projectile motion and the energy changes of a decelerating vehicle.

During an athletics competition, a javelin is thrown from a point O at ground level with an initial speed of 25 m/s at an angle of 40° above the horizontal. The javelin is assumed to be a particle and air resistance is ignored.

In a separate safety analysis, a car of mass 1,200 kg is moving along a straight horizontal road at 20 m/s. The driver applies the brakes, which produce a braking force of 4,800 N. The car also experiences a constant resistance force of 600 N throughout its motion.

Ms. Birungi's major objectives are to determine the flight characteristics of the javelin and to analyse the deceleration and energy dissipation of the braking car.

Task:

Help Ms. Birungi to:

a) For the javelin, find:
   (i) the total time of flight
   (ii) the maximum height reached above the ground
   (iii) the horizontal range

b) For the braking car, find:
   (i) the deceleration of the car
   (ii) the distance the car travels before coming to rest
   (iii) the total work done by the braking force

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